KINGDOM ANIMALIA

📝 Part 1: In-Chapter Point-Wise Notes 🦁✨

🧬 8.1 What is a Species?

- A **species** means an organism of a particular kind whose members can *interbreed* among themselves to produce fertile young ones.
- All individuals of a species can normally breed among themselves.
- Even if individuals have differences (like house cats varying in colour, height, or tail length), they can still interbreed and thus belong to the exact same species ➡️ *(Example: Felis domesticus)*.

🚫🔗 Crosses Between Different Species (Sterile Offspring)

- Interbreeding can sometimes occur between two *very closely related* species, but the offspring produced are invariably **sterile** (unable to reproduce).
- **Examples of Sterile Crosses:**
    - Horse + Donkey = **Mule**.
    - Zebra + Donkey = **Zenkey**.
    - Tiger + Lion = **Tigon**.
    - Goat + Sheep = **Geep**.
    - *Plant Example:* **Pomato** (produces potatoes underground and tomatoes above ground, but produces no seeds).

🐾 Kingdom Animalia Overview

- Organisms belonging to Kingdom Animalia are **multicellular, eukaryotic, and heterotrophic**.
- They obtain food by eating or sucking (differently nourished), and they exist entirely without a cell wall and without chlorophyll.
- They are usually mobile, though some specific exceptions like sponges and corals are non-mobile (sessile).

🐦‍⬛ The "Jungle Crow" Concept (Genus vs. Species)

- The hill crow (jungle crow) and the plain crow (house crow) differ in many features like size, colour, and beak.
- Because they cannot interbreed, they are classified as two **different species**.
- However, because they are very similar, both of these crows belong to the exact same **genus**, which is *Corvus*.

🌎 **Scientific Names (Binomial Nomenclature)**

- To entirely eliminate the confusion caused by variable local and common names, species are given a **two-part name** in scientific language.
- The first part of the name is the **genus**, and the second part is the name of the particular **species**.
    - ➡️ *(Example: The hill crow is Corvus macrorhynchos and the house crow is Corvus splendens).*
- This specific method of naming organisms is called **binomial nomenclature** (where *bi* means two, and *nomen* means name).
- It was first introduced by the scientist Linnaeus (1707-1778).

✍️ **Strict Rules for Writing Scientific Names:**

1. They are always written in the Roman script.
2. When printed in a book or typed, they are always written in *italics*.
3. **Crucial handwritten rule:** When hand-written, the names of the genus and the species **must be underlined separately**.
4. The first letter of the genus name must always be a **Capital letter**, while the species name must begin with a **small letter**.

📝 **Part 1: In-Chapter Point-Wise Notes**

📊 **A Sample Classification of Living Beings**

To understand the hierarchy from the highest level (Kingdom) to the lowest level (Species), here is how common organisms are classified:

Classification Level 📉	Modern Man 🧍	House Cat 🐈	Tiger 🐅	Dog 🐕	Mango 🥭
Kingdom	Animalia	Animalia	Animalia	Animalia	Plantae
Phylum	Chordata (Vertebrata)	Chordata (Vertebrata)	Chordata (Vertebrata)	Chordata (Vertebrata)	Spermatophyta
Class	Mammalia	Mammalia	Mammalia	Mammalia	Angiospermae
Order	Primates	Carnivora	Carnivora	Carnivora	Sapindales
Family	Hominidae	Felidae	Felidae	Canidae	Anacardiaceae
Genus	*Homo*	*Felis*	*Panthera*	*Canis*	*Mangifera*
Species	*sapiens*	*domesticus*	*tigris*	*familiaris*	*indica*

(Note: Additional common scientific names to memorize for ICSE: Honeybee = Apis indica, Housefly = Musca domestica, Peacock = Pavo cristatus, Cobra = Naja naja, Hoopoe = Upupa epops).

📌 8.6 & 8.7 Major Groups of Animals: Invertebrata vs Vertebrata

The animal kingdom is divided into nine major phyla. This classification is a matter of convenience based on the presence or absence of a backbone:

- **Invertebrates (Phyla 1-8):** Animals *without* a backbone. (Porifera, Cnidaria, Platyhelminthes, Nematoda, Annelida, Arthropoda, Mollusca, Echinodermata).

- **Vertebrates (Phylum 9):** *Animals with some kind of backbone (Chordata).*

🆚 Table 8.2: Major Differences Between Vertebrates and Invertebrates

Parameter 📏	Vertebrates 🦴	Invertebrates 🐛
Skeleton	They have an internal skeleton.	No internal skeleton.
Backbone	A backbone is present.	Backbone is not present.
Tail	A tail is usually present.	Tail is absent (anus at the tip of the back end of the body).
Heart Position	Heart is on the **ventral** (front) side of the body.	Heart, when present, is on the **dorsal** (back) side of the body.
Nerve Cord	Nerve (spinal) cord is **dorsal and hollow**.	Nerve cord is **ventral and solid**.
Limbs	They have two pairs of limbs.	They have three or more pairs of limbs, if present.
Blood	Haemoglobin is present in red blood cells.	Haemoglobin, if present, is dissolved directly in the blood plasma.

🥠 8.8 Invertebrate Phyla - Porifera to Echinodermata

🧬 Basis of Classification

Before diving into the phyla, it is crucial to understand the fundamental biological features used to classify these animals into different groups.

- **Level of Organization:** Ranges from basic to complex ➡️ Cellular level ➡️ Tissue level ➡️ Organ & Organ System level.

- **Segmentation:** Refers to the body being systematically divided into segments.
- **Germ Layers:** The foundational cell layers formed during embryonic development.
    - **Endoderm:** Inner layer.
    - **Mesoderm:** Middle layer.
    - **Ectoderm:** Outer layer.
    - **Diploblastic:** Organisms possessing only 2 germ layers.
    - **Triploblastic:** Organisms possessing all 3 germ layers.
- **Coelom:** A true body cavity (empty space) located inside the body, which must be strictly lined by the **Mesoderm**.

🧽 1. Phylum Porifera (Pore-bearing animals)

- **Organization:** Simplest multicellular animals exhibiting a **cellular level** of organization.
- **Body Symmetry:** Completely asymmetrical.
- **Habitat & Lifestyle:** Aquatic, fixed type (sessile) organisms that do not move.
- **Structure:** The body consists of a hollow tube. They lack a single mouth; instead, water enters through numerous tiny pores/canals called **Ostia** and exits through a single large opening on top called the **Osculum**.
- **Protection:** They possess a skeleton made of protective spongy fibres or microscopic spicules.
- ➡️ *(Examples: Sycon, bath sponge, sponges).*

🪸 2. Phylum Cnidaria / Coelenterata (Sac-like animals)

- **Organization:** Exhibits a **tissue level** of organization.
- **Body Symmetry:** Radially symmetrical.
- **Body Wall:** Possesses a two-layered body wall consisting of the **Epidermis** (outer layer) and **Gastrodermis** (inner digestive lining).

- **Defense & Feeding:** The epidermis contains specialized **stinging cells**. They possess tentacles to catch food organisms.
- **Digestive Cavity:** Encloses a single central cavity known as the **coelenteron**, which has only one opening (acting as a mouth) where digestion takes place.
- **Reproduction:** Bisexual organisms.
- **Special Feature:** Corals develop hard skeletons made up of calcium carbonate from their own secretions.
- ➡️ *(Examples: Hydra, jellyfish, sea anemone, corals).*

🪱 The Next Three Phyla - "The Worms"

These three phyla (Platyhelminthes, Nematoda, Annelida) include animals collectively called "worms"—a common term for crawling/creeping animals with relatively soft bodies.

🪱 3. Phylum Platyhelminthes (Flatworms)

- **Organization:** Exhibits an **organ & organ system level** of organization.
- **Body Symmetry:** Bilateral symmetry.
- **Coelom:** They are **Acoelomata** (body cavity is completely absent).
- **Structure:** Small, soft, flat, and usually unsegmented worms. *(Exception: the tapeworm is pseudosegmented).*
- **Digestive System:** Follows a blind sac body plan, meaning the alimentary canal has only **one opening**.
- **Reproduction:** Bisexual organisms.
- **Habitat:** Exist as free-living organisms in water or as parasites inside other animals.
- ➡️ *(Examples: Parasitic: Liver fluke, Tapeworm. Free-living: Planaria).*

🪱 4. Phylum Nematoda / Aschelminthes (Roundworms)

- **Organization:** Exhibits an **organ & organ system level** of organization.

- **Coelom:** Possesses a fluid-filled or **false body cavity** (pseudocoelom), thus lacking a true coelom.
- **Structure:** Soft, long, cylindrical, and strictly unsegmented body.
- **Digestive System:** Features a complete alimentary canal with **two openings** (mouth and anus).
- **Dimorphism:** Males and females look different (e.g., the male *Ascaris* is notably smaller than the female).
- **Habitat:** Mostly parasitic, though some live free in the soil.
- ➡️ *(Examples: Ascaris, hookworm, eel worm, filaria worm).*

🪱 5. Phylum Annelida (Segmented or ringed worms)

- **Organization:** Exhibits an **organ & organ system level** of organization.
- **Coelom:** They are **Coelomata** (a true body cavity is present between the body wall and alimentary canal).
- **Structure:** Soft, cylindrical body explicitly divided into distinct ring-like **segments**.
- **Digestive System:** A well-developed digestive system with an alimentary canal open at **two ends** (mouth and anus).
- **Excretion:** Excretion occurs through specialized tubular organs called **nephridia**.
- **Reproduction:** Bisexual organisms.
- ➡️ *(Examples: Earthworm, leech, Nereis).*

📝 Part 1: In-Chapter Point-Wise Notes

📐 Basis of Classification: Body Symmetry & Directions

Before classifying more complex animals, it is important to understand how their bodies are laid out.

- **Body Symmetry:**
    - **Asymmetry:** No matter how you cut the organism, the two halves will never be similar. ➡️ *(Example: Amoeba).*

- **Radial Symmetry:** Any line passing through the centre (along any radius) will divide the body into two similar halves. ➡️ *(Example: Starfish).*
- **Bilateral Symmetry:** The body can be divided into two similar halves *only along one plane* (the middle axis). Almost all vertebrates and a large number of invertebrates have this. ➡️ *(Example: Housefly, Human).*

- **Anatomical Body Directions:**
  - **Anterior:** Towards the front.
  - **Posterior:** Towards the hind end (behind).
  - **Dorsal:** Relating to the back or the upper side of the body (away from ground).
  - **Ventral:** Belonging to the lower or underside of the body (towards ground).
  - **Lateral:** Sides (right or left).

🪱 **The "Worms" Comparison Table (Platyhelminthes, Nematoda, Annelida)**

Understanding the evolution of the body cavity (coelom) is crucial here.

Feature 📏	**Flatworm (Platyhelminthes)**	**Roundworm (Nematoda)**	**Ringed worms (Annelida)**
Body Shape	Usually flat, ribbon-like, unsegmented.	Soft, round, cylindrical body.	Soft, cylindrical, ring-like segmented body.
Coelom (Body Cavity)	**Acoelomate:** Body cavity is absent.	**Pseudocoelomate:** False body cavity	**Coelomate:** True body

Feature 📏	Flatworm (Platyhelminthes)	Roundworm (Nematoda)	Ringed worms (Annelida)
		present (coelom in pouches).	cavity present.
Alimentary Canal	Has **one opening**, serving as both mouth and anus.	Has **two openings** (mouth and anus) at the two ends.	Opens at **two ends**.

🪱 **Deep Dive: The Earthworm (Phylum Annelida)**

- **Scientific Name:** *Pheretima posthuma* (Common Indian earthworm).
- **Habits:** They live in burrows in damp soil. They are **nocturnal** (avoid light) and crawl up to the surface at night for feeding and mating. Dry soil is fatal because their skin must be kept moist for respiration.
- **Structure:**
    - Long cylindrical body divided into **80-100 ring-like segments**.
    - No head, no appendages (legs), no eyes, and no tail.
    - The mouth is a simple opening at the front, and the anus is at the extreme hind end.
    - **Clitellum:** A short, cylindrical band of thick glandular skin surrounding segments 14-16 (about 1/3rd distance from the front). It serves in reproduction.
- **Reproduction:** They are **bisexual** (both male and female reproductive organs exist in one single body).
- **Economic Importance (Useful in Agriculture):**
    - Aerates and loosens the soil as it burrows, helping plant roots respire and penetrate deeper.
    - Interchanges topsoil with the lower layer, increasing fertility.

    - Excreta (castings) is rich in **nitrogenous matter (urea)** required for plant growth.
    - Used in vermiculture for high-quality manure and as bait for fishing.

🕷️ **6. Phylum Arthropoda (Jointed-legged animals)**

- **Scale:** This phylum contains the **largest number of species** in the entire animal kingdom.
- **Organization & Cavity:** They exhibit **bilateral symmetry** and are **Coelomata** (a true body cavity is present).
- **Body Division:** The body is distinctly divided into three parts — **head, thorax, and abdomen**.
- **Key Features:**
    - They have **jointed limbs/legs** (one pair on each or some body segments).
    - They possess a hard **exoskeleton made of chitin**.
- **Moulting:** Because the exoskeleton is hard, it breaks to provide space as the insect grows. The casting off and regrowing of this exoskeleton during early growth is collectively called **moulting**.
- ➡️ *(Examples: Crayfish, crabs, millipede, centipede, insects, scorpions, spiders).*
- *(Note: The further division of Arthropoda into classes like Crustacea, Myriapoda, Insecta, and Arachnida is beyond the syllabus and not required).*

🐌 **7. Phylum Mollusca (Soft-bodied animals)**

- **Organization & Cavity:** They are **triploblastic**, exhibit **bilateral symmetry**, and are **Coelomata** (true body cavity present).
- **Key Features:**
    - **Aquatic** animals with a soft, unsegmented body.
    - Most possess a hard, **calcareous shell** to protect the soft body.

    - They use specialized **gills for breathing**.
    - They possess a **muscular foot** for locomotion, which also acts as a weapon in some cases.
- ➡️ *(Examples from text and images: Mussel, chiton, Loligo (squid), snail, octopus, slug, Sepia (cuttlefish), oyster, clam).*

⭐ **8. Phylum Echinodermata (Spiny-skinned animals)**

- **Organization & Cavity:** They are **Coelomata** (true body cavity present) and exhibit **radial symmetry** (similar parts are arranged regularly around a central point).
- **Key Features:**
    - Strictly **marine animals** that are unsegmented.
    - The **head is absent**.
    - They typically have **five arms**.
    - They possess an **exoskeleton and a spiny surface**.
    - They move utilizing specialized structures called **tube-feet**.
- ➡️ *(Examples from text and images: Starfish, brittle-star, sea-urchin, sea cucumber).*

🦴 **9. Phylum Chordata (Vertebrata)**

- **The Notochord:** All chordates possess a notochord, which is a rod-like structure present in the mid-dorsal axis of the body.
    - In all vertebrates (including humans), the notochord is present during **embryonic life**.
    - Except for a few primitive forms, this notochord is eventually replaced by a **backbone (vertebral column)**.
- **Key Vertebrate Features:**
    - They possess a well-developed vertebral column forming the main axis of their internal skeleton, made of bones or cartilage.

    - The body is divided into a **head, a trunk, and two pairs of appendages**.
    - **Gill slits** are present at some time in their life cycle.
- **Five Classes of Vertebrates:** The vertebrates are divided into five classes comprising fishes, frogs (Amphibians), lizards (Reptiles), birds (Aves), and hairy quadrupeds (Mammals).

🪱 **Invertebrates**

- **Nerve Cord:** Present on the **ventral** side.
- **Heart:** Located on the **dorsal** side.
- **Blood:** Haemoglobin (if present) is **dissolved directly in the blood** plasma.

🦴 **Vertebrates (Phylum Chordata)**

- **Skeleton: Notochord / vertebral column** is definitively present.
- **Heart:** Located on the **ventral** side.
- **Nerve Cord:** Present on the **dorsal** side.
- **Blood:** Haemoglobin is contained strictly within **RBCs (Red Blood Cells)**.
- ➡️ *(Examples: Fishes, birds, humans).*

📝 **Part 1: In-Chapter Point-Wise Notes (Restored Pattern)** 🦁✨

🐟 **8.9.1 Vertebrate Classes: (i) Class Pisces (Fishes)**

- **Habitat & Lifestyle:** Completely adapted to aquatic life.
- **Temperature Regulation:** Cold-blooded or **poikilothermal** (their body temperature fluctuates directly with the temperature of the environment).
- **Key Features:** Body is covered with scales and slimy skin. They locomote utilizing fins and have absolutely no limbs.
- **Respiration:** They breathe by means of gills.
- **Heart Structure:** They possess a **2-chambered heart**.
- **Reproduction:** Fertilization is strictly **external**.

- **Sub-categories of Fishes:**
    - **Cartilaginous fishes:** Skeleton is made entirely of soft cartilage. Their gills are exposed without a cover. ➡️ *(Examples: Shark, dogfish, skates).*
    - **Bony fishes:** Skeleton is made of bones. Their gills are safely concealed under a gill cover called an **operculum**. ➡️ *(Examples: Trout, rohu, carps, roaches, herrings).*
- ➡️ *(Master Example List: Shark, dogfish, skates, trout, rohu, tuna, carps, roaches, herrings).*

🐸 **(ii) Class Amphibia**

- **Habitat & Lifestyle:** They lead a dual life. They spend a major part of their adult lives on land, while their egg and larval stages are aquatic.
- **Temperature Regulation:** Cold-blooded (**poikilothermal**).
- **Key Features:** They have smooth, non-scaly, moist, and slimy skin. They possess 2 pairs of limbs that are five-fingered (pentadactyl). The eardrum (tympanum) lies directly on the surface of the skin.
- **Respiration:** Larvae breathe by means of gills, whereas adult amphibians breathe by means of lungs and skin.
- **Heart Structure:** They possess a **3-chambered heart**.
- **Reproduction:** Eggs are laid in water, and fertilization is **external**.
- ➡️ *(Master Example List: Frog, toad, midwife toad, tree frog, salamander, newt).*

🦎 **(iii) Class Reptilia**

- **Habitat & Lifestyle:** Completely adapted to life as land animals.
- **Temperature Regulation:** Cold-blooded (**poikilothermal**).
- **Key Features:** They have rough, dry skin covered with horny scales. They possess two pairs of limbs. The eardrum (tympanum) lies at the bottom of a tubular depression or pit.
- **Respiration:** They breathe by means of lungs right from birth.

- **Heart Structure:** They possess a **3-chambered heart** in which the ventricle is partially divided.
    - **Crucial Exception:** The crocodile has a fully **4-chambered heart**.
- **Reproduction:** They lay eggs that have a **leathery shell**.
- ➡️ *(Master Example List: Lizard, snake, tortoise, turtle, crocodile, alligator, gharial—found only in the rivers of India, Malaysia, and Myanmar).*

🦅 **(iv) Class Aves (Birds)**

- **Habitat & Lifestyle:** They are aerial animals.
- **Temperature Regulation:** Warm-blooded or **homeothermal** (they maintain an almost constant, relatively high body temperature irrespective of the environment).
- **Key Features:** Their body is streamlined and the skeleton is light to facilitate flying. Their forelimbs are modified into wings. They have an exoskeleton of feathers and scaly legs. Interestingly, **both skeletons** (endo and exo) are present.
- **Respiration:** They breathe by means of lungs.
- **Heart Structure:** They possess a **4-chambered heart**.
- **Reproduction:** They lay eggs with **calcareous shells**.
- ➡️ *(Master Example List: Pigeon, sparrow, crow, duck, owl, penguin, ostrich, emu, common myna, sunbird, hoopoe, pidda, pond heron, parrot).*

🐅 **(v) Class Mammalia (Hairy quadrupeds)**

- **Habitat & Lifestyle:** The most highly developed animals for life on land, living in all sorts of habitats. *(Exceptions: Whales, dolphins, and seals live in water; bats fly in the air).*
- **Temperature Regulation:** Warm-blooded (**homeothermal**).
- **Key Features:** They have **hair** on their skin. They have a muscular diaphragm separating the thorax and abdomen internally. They possess a

pair of external ears (pinnae) and external testes contained in a scrotum. They possess two pairs of limbs.

- **Respiration:** They breathe by means of lungs.
- **Heart Structure:** They possess a **4-chambered heart**.
- **Reproduction:** They are **viviparous** (give birth to living young ones) and nourish their young by means of milk-secreting **mammary glands**.
    - **Crucial Exceptions (Egg-Laying Mammals):** The **Australian spiny ant-eater** and the **duckbilled platypus** are exceptions which lay eggs.
    - *(Note: Do not confuse the Spiny ant-eater of Australia with the hedgehog. The hedgehog gives birth to young ones).*
- ➡️ *(Master Example List: Cat, dog, cow, sheep, rat, bat, seal, monkey, apes, man, whale, dolphin).*

💡 Do You Know? (Extra Fascinating Facts)

The Cuckoo ("Koel") & Brood Parasitism 🐓 It is the black coloured male that sings the famous *(koo-oo-koo-oo)*, while the brownish white-patched female produces only a dull broken call *(kuk ... kuk ...)*. Instead of building their own nest, the female lays her eggs in a **crow's nest**, and the crow unknowingly incubates them. This clever survival strategy, where a bird relies entirely on other birds to raise its young ones, is known scientifically as **brood parasitism**.

The Extremes of the Ostrich 🥚 The ostrich is the largest and heaviest living bird. Although it is completely flightless, it possesses incredibly powerful legs for running, making it the **fastest bird on land**.

- It lays the largest of all bird eggs.
- Its eyes are the largest of any land animal—in fact, they occupy so much space within the skull that the **brain of an ostrich is actually much smaller than either one of its eyeballs!**

🏆 Part 2: End-of-Chapter Master Lists (Division 1)

Focus: Terminology & The Ultimate Example Bank 🌍🐾

📚 1. Combined Terms List (Glossary)

Biological Term 🔑	Exam-Ready Definition 📝
Species	An organism of a particular kind whose members can interbreed among themselves to produce fertile young ones.
Binomial Nomenclature	The scientific method of naming organisms using a two-part name (Genus + species) introduced by Linnaeus.
Invertebrate	An animal that does not possess a backbone or vertebral column.
Vertebrate	An animal (Chordate) that possesses a notochord or a well-developed vertebral column (backbone).
Coelom	A true body cavity (empty space) located inside the body that is strictly lined by the mesoderm.
Acoelomate	Organisms entirely lacking a true body cavity (e.g., Flatworms).
Pseudocoelomate	Organisms possessing a "false" or fluid-filled body cavity in pouches (e.g., Roundworms).
Moulting	The periodic casting off and regrowing of the rigid exoskeleton in arthropods to allow for body growth.
Poikilothermal (Cold-blooded)	Animals whose body temperature fluctuates directly with the temperature of their environment (e.g., Fishes, Amphibians, Reptiles).

Biological Term 🔑	Exam-Ready Definition 📝
Homeothermal (Warm-blooded)	Animals that maintain an almost constant, relatively high body temperature irrespective of the environment (e.g., Birds, Mammals).
Brood Parasitism	A survival strategy where an organism (like the Cuckoo) relies entirely on other species (like the Crow) to incubate its eggs and raise its young.

🌟 2. Taxonomic Master Example List

Phylum / Class 🧬	Key Exam Examples to Memorize 🎯
Porifera	Sycon, Bath sponge.
Cnidaria (Coelenterata)	Hydra, Jellyfish, Sea anemone, Red Coral.
Platyhelminthes	Tapeworm, Liver fluke (Parasites) | Planaria (Free-living).
Nematoda	*Ascaris*, Hookworm, Eelworm, Filaria worm.
Annelida	Earthworm (*Pheretima posthuma*), Leech, Nereis.
Arthropoda	Crayfish, Crabs, Millipede, Centipede, Scorpion, Spider, Insect (Housefly, Honeybee).
Mollusca	Snail, Slug, Oyster, Mussel, Clam, Octopus, Chiton, *Loligo* (Squid), *Sepia* (Cuttlefish).
Echinodermata	Starfish, Brittle-star, Sea-urchin, Sea cucumber.

Phylum / Class 🧬	Key Exam Examples to Memorize 🎯
Pisces (Cartilaginous)	Shark, Dogfish, Skates.
Pisces (Bony)	Trout, Rohu, Carps, Roaches, Herrings, Tuna.
Amphibia	Frog, Toad, Midwife toad, Tree frog, Salamander, Newt.
Reptilia	Lizard, Snake (Cobra), Tortoise, Turtle, Crocodile, Alligator, Gharial.
Aves	Pigeon, Sparrow, Crow, Duck, Owl, Penguin, Ostrich, Emu, Sunbird, Hoopoe, Parrot.
Mammalia	Human (*Homo sapiens*), Cat, Dog, Cow, Sheep, Rat, Bat, Seal, Monkey, Whale, Dolphin.

🏆 Part 2: End-of-Chapter Master Lists (Division 2)

Focus: Crucial Exceptions & Function Matrix ⚙️🔍

🚨 3. The "Ultimate Exceptions" List (Examiners Love These!)

- **The Segmented Exception:** Almost all flatworms (Platyhelminthes) are unsegmented, *except* the **Tapeworm**, which is pseudosegmented.
- **The Size Exception:** In Nematodes (like *Ascaris*), the **male is significantly smaller** than the female.
- **The Heart Exception:** All reptiles have a 3-chambered heart with a partially divided ventricle, *except* the **Crocodile**, which has a fully **4-chambered heart**.
- **The Egg-Laying Mammal Exceptions:** All mammals give birth to live young (viviparous), *except* the **Australian spiny ant-eater** and the **Duckbilled platypus**, which lay eggs.

- **The Habitat Exceptions in Mammals:** Mammals are primarily land animals, *except* **whales and dolphins** (fully aquatic) and **bats** (aerial/flying).
- **The Brain/Eye Exception:** The **Ostrich's** eyes are so massively large that its brain is actually *smaller* than either one of its eyeballs.
- **The Sterile Offspring Exceptions:** Mating between two closely related but different species results in sterile young ➡️ Horse + Donkey = **Mule**; Zebra + Donkey = **Zenkey**; Tiger + Lion = **Tigon**.

🛠️ 4. Master Function & Location Matrix

Anatomical Structure ⚙️	**Associated Phylum/Class** 📍	**Primary Function / Fact** 🎯
Ostia / Osculum	Porifera	**Ostia:** Tiny pores for water entry. **Osculum:** Large top opening for water exit.
Epidermis 🛡️	Cnidaria	The protective outer layer containing specialized **stinging cells** to zap prey! ⚡
Gastrodermis 🍕	Cnidaria	The inner lining acting as a "stomach skin" to digest food inside the coelenteron. 😋
Clitellum	Annelida (Earthworm)	A thick glandular band (segments 14-16) that serves a critical role in reproduction.
Nephridia	Annelida	Specialized tubular organs used strictly for **excretion**.

Anatomical Structure ⚙️	Associated Phylum/Class 📍	Primary Function / Fact 🎯
Chitinous Exoskeleton	Arthropoda	Provides a hard outer protective shell that must be periodically shed (moulted) for growth.
Muscular Foot	Mollusca	Aids in locomotion and can also act as a weapon in some cases.
Tube-Feet	Echinodermata	Specialized structures utilized for movement across the sea floor.
Operculum	Pisces (Bony Fishes)	A protective bony flap that entirely conceals and covers the gills.
Tympanum (Eardrum)	Amphibians & Reptiles	Detects sound. (In amphibians it sits on the skin surface; in reptiles, it is in a tubular pit).
Diaphragm	Mammalia	A muscular partition internally separating the thorax and the abdomen.
Mammary Glands	Mammalia	Secrete milk to nourish live-born young ones.

🟢 **Progress Check (Page 84)**

1. Rearrange the categories in a sequence from highest downward to lowest: 📉

- Kingdom ➡️ Phylum ➡️ Class ➡️ Genus ➡️ Species

2. **Tick-mark the features that characterise vertebrates:** ✔️

- **(i) Dorsal nerve cord** ✅ *(Correct! Vertebrates have a hollow dorsal nerve cord).*
- (ii) Haemoglobin dissolved ❌ *(Incorrect. It is contained within Red Blood Cells).*
- (iii) Three or more pairs of limbs ❌ *(Incorrect. They typically have two pairs).*
- **(iv) A tail behind the level of the anus** ✅ *(Correct! A tail is usually present).*
- (v) Dorsal heart ❌ *(Incorrect. Vertebrates have a ventrally located heart).*

🟢 **Progress Check (Page 89)**

1. **Mention the name of the phylum against the following characteristics:** 📝

- (i) No single mouth but numerous pores as water inlets ➡️ **Porifera** 🧽
- (ii) Jointed appendages ➡️ **Arthropoda** 🕷️
- (iii) Unsegmented body with a protective calcareous shell ➡️ **Mollusca** 🐌
- (iv) Locomotion by tube feet ➡️ **Echinodermata** ⭐
- (v) Long, cylindrical and unsegmented body ➡️ **Nematoda** 🪱
- (vi) Small flat unsegmented worms ➡️ **Platyhelminthes** 🪱

2. **Name the phylum of each of the following invertebrates:** 🏷️

- (i) Leech ➡️ **Annelida**
- (ii) Hydra ➡️ **Cnidaria (Coelenterata)**
- (iii) Tapeworm ➡️ **Platyhelminthes**
- (iv) Sea cucumber ➡️ **Echinodermata**
- (v) Sponge ➡️ **Porifera**
- (vi) Roundworm ➡️ **Nematoda**
- (vii) Scorpion ➡️ **Arthropoda**

- (viii) Slug ➡️ **Mollusca**
- (ix) Centipede ➡️ **Arthropoda**
- (x) Butterfly ➡️ **Arthropoda**

🟢 **Progress Check (Page 92)**

1. **Match characteristics (Column I) with the classes (Column II):** 🧩
   - **(a) Pisces:** (iii) Two-chambered heart, (v) Scales, (vii) Gills. 🐟
   - **(b) Amphibia:** (i) Three chambered heart, (viii) Moist skin. 🐸
   - **(c) Reptilia:** (i) Three chambered heart, (iv) Eardrum in a pit, (ix) Horny scales on skin. 🦎
   - **(d) Aves:** (ii) Feathers, (x) Homeothermal. 🪶
   - **(e) Mammalia:** (vi) External ears, (x) Homeothermal, (xi) Testes in scrotum. 🐅
2. **Mention the class of vertebrates against each:** 🐾
   - (i) Crocodile ➡️ **Reptilia**
   - (ii) Penguin ➡️ **Aves**
   - (iii) Toad ➡️ **Amphibia**
   - (iv) Shark ➡️ **Pisces**
   - (v) Whale ➡️ **Mammalia**
   - (vi) Bat ➡️ **Mammalia**
   - (vii) House lizard ➡️ **Reptilia**

📘 **Review Questions: A. MULTIPLE CHOICE TYPE**

- **(i)** Kingdom Animalia are mainly grouped under two categories: ➡️ **(d) vertebrates and invertebrates.** 🦴🦎
- **(ii)** *Note: This book question is flawed as every single option contains an odd-one-out that does not belong to the stated group! For your knowledge:*
  - (a) Fern is a plant, not Fungi. 🌿

- (b) Sponge is an animal, not Protista. 🧽
- (c) Starfish & Roaches are invertebrates, not Pisces. ⭐
- (d) Oyster is an invertebrate, not a Vertebrate. 🦪

- **(iii)** Example of a binomial scientific name: ➡️ **(c) *Rana tigrina*.** 🐸
- **(iv)** Correct statement about the respective animals: ➡️ **(d) Mule is neither a donkey nor a horse.** 🐴
- **(v)** Lowest category among the levels of classification (from the options): ➡️ **(d) Genus.** 📉
- **(vi)** Unicellular, prokaryotic organisms belong to: ➡️ **(c) Monera.** 🦠
- **(vii)** Which of the following organisms is chlorophyllous? ➡️ **(b) Euglena.** 🌞
- **(viii)** Which of the following belongs to class Pisces? ➡️ **(b) Dogfish.** 🦈
- **(ix)** Identify the invertebrate phyla (P: Spicules, Q: False body cavity): ➡️ **(c) P – Porifera, Q – Nematoda.** 🧽🪱

📕 **Review Questions: B. VERY SHORT ANSWER TYPE**

1. Who had introduced the binomial system of naming living beings?

- **Linnaeus.** 👨‍🔬

2. Which two characters are common to dog, humans, squirrel, bat, camel and monkey?

- **(c) external ears** AND **(d) give birth to young ones.** 🐿️🦇

3. Match the items in Column I with as many as possible from Column II: 🔗

- 1. Pine ➡️ **(iv) Gymnosperm, (v) Plantae**
- 2. Earthworm ➡️ **(vi) Animalia**
-

3. Bread mould ➡️ **(i) Fungi**

-

4. Amoeba ➡️ **(vii) Protista**

-

5. Moss ➡️ **(viii) Bryophyta, (v) Plantae**

-

6. Bacteria ➡️ **(ii) Monera, (iii) Prokaryote**

4. What are the group names for the following categories of animals? 🤌

- (a) Animals with a backbone ➡️ **Vertebrates (Chordata)**
- (b) Animals with a hairy skin ➡️ **Mammals (Mammalia)**
- (c) Animals with three pairs of legs ➡️ **Insects (Insecta)**
- (d) Animals with feathers ➡️ **Birds (Aves)**

5. Which ones of the following animals are invertebrates? 🐛

- **Housefly, silverfish, jellyfish, and sponge.** 🤌

6. Name three animals (belonging to different classes) which breathe by means of lungs but have no external ears (pinnae). 🫁

- **Frog** (Amphibia), **Lizard/Snake** (Reptilia), and **Pigeon/Crow** (Aves). 🐸🦎🦅

7. Underline the term which does NOT match the animal: 🚫

- (i) Amoeba – Nucleus, **tentacle**, food vacuole.
- (ii) Hydra – Invertebrata, Cnidaria, **Crustacea**.
- (iii) Fish – Gills, paired fins, **ear drum**.
- (iv) Earthworm – Invertebrata, Annelida, **Insecta**.
- (v) Grasshopper – Wings, trachea, **proboscis**.
- (vi) Butterfly – Insecta, Invertebrata, **Mollusca**.
- (vii) Whale – **Gills**, mammary glands, fat under the skin.

- (viii) Pigeon – Feathers, wings, **hair**.
- (ix) Monkey – External ear, sweat glands, **lateral line**.
- (x) Bat – **Aves**, Mammalia, Chordata.

📝 C. SHORT ANSWER TYPE

- **1. Name these:**
  - **(a)** The five kingdoms under the recent scheme of classification are Monera, Protista, Fungi, Plantae, and Animalia.
  - **(b)** The eight phyla of Invertebrates are Porifera, Cnidaria, Platyhelminthes, Nematoda, Annelida, Arthropoda, Mollusca, and Echinodermata.
  - **(c)** The five classes of Chordata are Pisces, Amphibia, Reptilia, Aves, and Mammalia.
  - **(d)** The two types of fishes on the basis of their skeletons are Cartilaginous fishes and Bony fishes.
- **2. Define the terms:**
  - **(a) Species:** An organism of a particular kind whose members can interbreed among themselves to produce fertile young ones.
  - **(b) Clitellum:** A short cylindrical band of thick glandular skin surrounding the earthworm's body (segments 14-16) that serves in reproduction.
  - **(c) Moulting:** The casting off and regrowing of the rigid exoskeleton in arthropods to provide space for body growth.
  - **(d) Diaphragm:** A muscular partition internally separating the thorax and the abdomen in mammals.
  - **(e) Nocturnal animals:** Animals that are active during the night and avoid light during the daytime.
- **3. Write the specific location and function of:**

- **(a) Microscopic spicules or elastic spongin fibre:** Location is the body wall of Porifera (sponges). Function is to provide a protective skeleton.
- **(b) Tentacles:** Location is surrounding the mouth in Cnidarians. Function is to catch food organisms.
- **(c) Parapodium:** Location is on the lateral sides of body segments in certain annelids like Nereis. Function is locomotion and respiration.
- **(d) Fins:** Location is on the exterior body of fishes (Pisces). Function is locomotion in aquatic environments.
- **(e) Feathers:** Location is covering the body of birds (Aves). Function is an exoskeleton that facilitates flying and provides insulation.

- **4. Rearrange categories starting from the highest:** Phylum ➡️ Class ➡️ Order ➡️ Family ➡️ Genus ➡️ Species.
- **5. Match Column I with Column II:**
  - Annelida ➡️ Earthworm.
  - Porifera ➡️ Sponge.
  - Mollusca ➡️ Octopus.
  - Reptilia ➡️ Snake.
  - Pisces ➡️ Trout.
  - Mammal ➡️ Rabbit.
  - **Left out:** "Amphibia" (Group) and "Pigeon" (Animal) do not match, because Pigeon belongs to Class Aves, and there is no amphibian listed in Column II.
- **6. Why are scientific names better?** Because common names are highly variable and confusing across different languages and localities. Scientific names utilize binomial nomenclature to give a universal, standardized two-part name that eliminates all confusion.

- **7. Two drawbacks of the old two-kingdom classification:** It grouped highly distinct organisms (like bacteria and fungi) together with plants. It also completely failed to distinguish between prokaryotic and eukaryotic cellular structures.
- **8. Why humans belong to *Homo sapiens* despite differences:** Because all individuals of this species can normally interbreed among themselves to produce fertile young ones, regardless of variations in facial features, colour, or height.
- **9. One difference between:**
    - **(a) Insecta and Arachnida:** Insects possess exactly three pairs of legs, whereas arachnids possess four pairs of legs.
    - **(b) Flatworm and Roundworm:** Flatworms are acoelomate (lacking a body cavity), whereas roundworms are pseudocoelomate (possessing a false body cavity).
- **10. One major similarity and difference:**
    - **(a) insects and birds:** Similarity is that both are aerial and capable of flight. Difference is insects are invertebrates (Arthropoda) while birds are vertebrates (Aves).
    - **(b) whales and fishes:** Similarity is both are completely adapted to aquatic life. Difference is whales are warm-blooded, lung-breathing mammals, whereas fishes are cold-blooded, gill-breathing Pisces.
    - **(c) snakes and earthworms:** Similarity is both have an elongated, tubular body without limbs. Difference is snakes are vertebrates (Reptilia) while earthworms are invertebrates (Annelida).
    - **(d) bat and pigeon:** Similarity is both are warm-blooded, aerial creatures. Difference is bats are viviparous mammals with hair, while pigeons are egg-laying birds with feathers.
    - **(e) cuttlefish and dogfish:** Similarity is both are strictly marine aquatic animals. Difference is cuttlefish is a soft-bodied invertebrate (Mollusca), whereas dogfish is a cartilaginous vertebrate (Pisces).

- **(f) wall lizard and frog:** Similarity is both are cold-blooded (poikilothermal). Difference is wall lizards have dry, horny scales (Reptilia), whereas frogs possess smooth, non-scaly, moist skin (Amphibia).

📘 D. LONG ANSWER TYPE

- **1. Distinguish between:**
  - **(a) Invertebrates and vertebrates:** Invertebrates lack a backbone, possess a ventral nerve cord, and have a dorsal heart. Vertebrates strictly possess a backbone (or notochord), a dorsal hollow nerve cord, and a ventral heart.
  - **(b) Ectothermal and Homeothermal:** Ectothermal (cold-blooded) animals have body temperatures that fluctuate directly with the environmental temperature. Homeothermal (warm-blooded) animals maintain a nearly constant, high internal body temperature irrespective of the environment.
  - **(c) Monera and Protista:** Monerans are strictly prokaryotic and unicellular. Protistans are eukaryotic and predominantly unicellular.
  - **(d) Roundworms and Ringed worms:** Roundworms (Nematoda) are unsegmented and possess a pseudocoelom (false body cavity). Ringed worms (Annelida) are distinctly segmented and possess a true coelom.
- **2. Cold-blooded vs Warm-blooded:** Cold-blooded organisms cannot regulate their internal body heat, causing their temperature to match their surroundings. Warm-blooded organisms possess internal physiological mechanisms to sustain a constant body temperature.
- **3. Whale Classification (HOTS):** A whale is classified under Mammalia because it gives birth to living young ones (viviparous) rather than laying eggs. Additionally, it nourishes its young with milk from mammary glands and breathes via lungs, not gills.
- **4. Birds Physics Link (SI):** Low bone density drastically reduces the bird's overall mass, which minimizes the gravitational pull (Weight = mass x gravity) acting upon them, making lift much easier. A streamlined body

shape powerfully reduces air resistance (drag) as they move forward, enabling aerodynamic efficiency.

🎨 E. STRUCTURED / APPLICATION TYPE

- **1. Peacock Image:**
  - **(a)** Kingdom: Animalia. Phylum: Chordata. Class: Aves.
  - **(b)** They are warm-blooded (homeothermal) and their forelimbs are structurally modified into wings.
  - **(c)** Phylum Chordata features a notochord and a dorsal nerve cord. It is divided into Pisces, Amphibia, Reptilia, Aves, and Mammalia.
- **2. Figures A, B, C, D:**
  - **(a)** Phylum: Cnidaria (Coelenterata).
  - **(b)** Habitat: Predominantly aquatic and mostly marine.
  - **(c)** (A) Sea Anemone, (B) Hydra, (C) Jellyfish, (D) Coral.
  - **(d)** Cavity name: Coelenteron.
  - **(e)** Locomotory structure: Tentacles.
- **3. Figure of the Frog:**
  - **(a)** Phylum: Chordata. Class: Amphibia.
  - **(b)** Larval stage uses gills; adult stage primarily uses lungs and skin.
  - **(c)** Structure (A) is the Eardrum (Tympanum). Its function is to detect sound waves.
  - **(d)** Label (B) is the Fore limb. Label (C) is the Hind limb.
  - **(e)** This animal is cold-blooded (poikilothermal).

😳 F. ASSERTION - REASON TYPE QUESTIONS

- **(i)** Answer: **(b)**. Both Assertion (similar as both are unicellular) and Reason (differ in nuclei) are true facts, but the Reason is not the correct explanation for the Assertion.

- **(ii)** Answer: **(a)**. Both are true, and the Reason perfectly explains why gymnosperms are considered to bear naked seeds.
- **(iii)** Answer: **(c)**. The Assertion is true (they cannot breed), but the Reason is entirely false because two different species definitively cannot interbreed.
- **(iv)** Answer: **(b)**. Both statements about Ostia and Osculum are factually true, but the definition of the Osculum is not the explanation for why Ostia are the entry openings.
- **(v)** Answer: **(d)**. The Assertion is false because the clitellum is located at the anterior portion (segments 14-16), not the posterior end. However, the Reason (it serves in reproduction) is perfectly true.
- **(vi)** Answer: **(c)**. The Assertion is true (Pisces have 2 chambers, Amphibians have 3 chambers), but the Reason provides completely false numbers (claiming 1-chamber and 2-chamber respectively).

🌍 G. CASE STUDY BASED QUESTIONS

- **1. Manas's Compost Pit:**
  - **(a)** The animal is an Earthworm. It belongs to Phylum Annelida.
  - **(b)** Two more organisms from this phylum are the Leech and Nereis.
  - **(c)** Two other phyla containing "worms" are Platyhelminthes (flatworms) and Nematoda (roundworms).
  - **(d)** They respire entirely through their moist, slimy skin.
  - **(e)** 'Nocturnal' means the animals are highly active during the night and completely avoid light during the daytime.
- **2. Coral Reefs & SDGs:**
  - Chemical changes like increased carbon dioxide drastically raise the acidity of ocean water. This acidity dissolves the hard calcium carbonate skeletons built by the tiny Cnidarian animals, destroying the vital habitat that supports immense marine biodiversity. This massive damage to the reefs severely hurts local tourism, diving,

and fishing industries, which directly devastates human livelihoods and local economies.

🦖 TYRANNOSAURUS CLASSIFICATION BOX

- **What do you think about its classification?** It was classified as a **Reptile**.
- **Can you call it a relative of Kangaroo? No**.
- **Give reason:** Kangaroos are classified as Mammals because they have hair and give birth to young ones, whereas Tyrannosaurus belongs to Class Reptilia. They are entirely different classes of vertebrates! 🌟